Retrieval-Augmented Memory for Long-Horizon Video World Models

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1. Introduction

One of the central challenges in robotics is operating under partial observability: an agent only sees a small part of the environment at any given moment, but still has to act as if it understands the larger scene. Computer vision is a key part of solving this problem because visual observations provide the evidence from which the agent must infer, update, and remember the structure of the surroundings over time. In many settings, especially embodied and interactive ones, a system has to reason over time. It must track what has already been seen, maintain a coherent representation of the environment, and use that representation to predict future observations. This is difficult because visual input arrives sequentially, and a useful model therefore needs not only perception but also memory.

World models are one framework for addressing this problem. Broadly speaking, a world model learns a compact representation of an environment and uses it to predict how observations evolve over time. This idea has been influential in both vision and reinforcement learning. Previous work such as PlaNet and Dreamer showed that latent dynamics models can support prediction and control directly from visual observations [3,4]. More recent methods, including VideoGPT and IRIS, demonstrated that latent-token and transformer-based architectures can model visual sequences effectively [6, 8]. Together, these results suggest that predictive latent modeling is generally a promising direction for embodied visual intelligence.

At the same time, long-horizon consistency remains a clear weakness of current approaches. A model may generate plausible short-term predictions while still failing to preserve a stable representation of a scene over longer trajectories. Once earlier observations fall outside the active context window, the model can start drifting (object identities may change, spatial layout may become inconsistent, and previously seen scenes may be replaced by something merely plausible rather than correct). Transformer-based world models have improved sequential prediction [1], but finite context still limits how well they handle information that becomes relevant again only after a long gap. Recent work such as WorldMem argues that explicit memory may be necessary for sustaining long-term visual consistency [7].

This issue becomes particularly acute in indoor embodied environments. An agent may observe a room, move through other parts of the scene, and later return from a different viewpoint. In that case, a robust world model should still preserve the identity of the original room: its rough layout, its objects, and their arrangement. If the model instead alters furniture or predicts a different but still reasonable-looking room, then it is not really modeling a persistent environment. This leads to the main question we study in this project: can a world model maintain scene consistency over time, even when the agent revisits parts of the environment after a long gap?

We study this question using AI2-THOR and ProcTHOR, which provide controllable indoor environments, embodied actions, and structured scene metadata [2, 5]. These simulators are well suited to our setting because they allow us to generate trajectories and to evaluate consistency in a controlled way. However, our project is intentionally narrow in scope, as we are not trying to train a large general-purpose world model across many homes or room categories. Instead, we focus on a small proof-of-concept setting, training a compact model centered on just a few rooms, and testing whether it can preserve one room across revisit. This makes our project feasible while still isolating the memory problem we care about.

To address this problem, we propose a retrieval-augmented latent world model for indoor revisit. The model first encodes visual observations into compact latent representations. A small transformer-based dynamics model then predicts future latent states from recent observations and actions. We augment this baseline with an external memory bank that stores selected past latent states together with simple metadata such as timestep or camera position. During rollout, the model can retrieve relevant earlier states and condition its predictions on them in addition to the recent context, and if information about the room has dropped out of the local context window, targeted retrieval may recover it more effectively than by relying on the sequence model only.

The project's contribution is an investigation of whether the model can stay stable over a long simulation period. First, we estimate long-horizon consistency through a revisit, which gives a more direct test of memory than standard short-horizon prediction metrics alone. Second, we examine whether a lightweight retrieval mechanism can improve consistency in a setting that is realistic under course-project compute limits. Third, we evaluate the model using metrics that better suit the question at hand, including visual similarity at revisit time, object persistence, and coarse agreement with the original room layout.

Because this is an introduction draft, our results are currently prospective rather than final. We expect retrieval-based memory to matter most as the temporal gap between first observation and revisit increases. For short trajectories, recent context may already be sufficient. For longer ones, however, earlier room information is more likely to be forgotten, and explicit retrieval should help reduce drift and preserve scene identity more reliably.

Overall, this project studies a small but meaningful version of a broader computer vision problem: how to maintain a consistent visual model of an environment over time. By narrowing the task to one-room revisit, we aim to test whether explicit memory improves long-horizon prediction in a controlled embodied setting.

2. AI Use

Generative AI tools were not used in the preparation of this paper introduction.

References

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